

EE3621: Digital Signal Processing

Lecture 22: Design of FIR filter - windowing method (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** The Core Philosophy of the Windowing Method: The Inverse DTFT dilemma—ideal frequency-selective filters possess non-causal, infinitely long impulse responses ($h_d[n] \sim \text{sinc}(n)$). Direct truncation via windowing ($h[n] = h_d[n] \cdot w[n]$) and the dual operation in the frequency domain: periodic circular convolution $H(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) * W(e^{j\omega})$.
 - **06:00 – 15:00 (9 mins):** The Gibbs Phenomenon & Spectral Leakage: Abrupt rectangular truncation, the Dirichlet sinc kernel $W_R(e^{j\omega}) = \frac{\sin(\omega N/2)}{\sin(\omega/2)}$, proof of the invariant 8.95% peak overshoot (1.0895 or ≈ 0.75 dB ripple), and why increasing filter length N only compresses ripple width without reducing overshoot amplitude.
 - **15:00 – 24:00 (9 mins):** Mathematical Formulation of Classic Window Families: Rectangular, Bartlett (Triangular), Hann (Raised Cosine), Hamming (Shifted Raised Cosine), and Blackman. Derivations of mainlobe widths ($\Delta\omega = 4\pi/N, 8\pi/N, 12\pi/N$), peak side-lobe levels ($-13, -25, -31, -41, -57$ dB), and minimum stopband attenuation A_s .
 - **24:00 – 30:00 (6 mins):** Formulation of Causal Ideal Impulse Responses: Analytical expressions for Ideal Lowpass, Highpass, Bandpass, and Bandstop filters with causal linear-phase delay $\tau = (N - 1)/2$. Resolving the indeterminate $0/0$ form at the center of symmetry.
 - **30:00 – 38:00 (8 mins):** Step-by-Step Worked Numerical Examples: Complete design of a 7-tap Lowpass filter using a Hann window, design of a 9-tap Highpass filter comparing Rectangular vs. Hamming truncation, and design of an 11-tap Bandpass filter using a Bartlett window.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
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1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Explain** the mathematical necessity of windowing in FIR filter design from an Inverse DTFT perspective.

2. **Analyze** the frequency-domain mechanism of windowing as a periodic convolution between the ideal brickwall response $H_d(e^{j\omega})$ and the window spectrum $W(e^{j\omega})$.
3. **Describe** the Gibbs phenomenon, proving why the peak overshoot of approximately 8.95% does not vanish as filter length $N \rightarrow \infty$ for a rectangular window.
4. **Quantify** the fundamental trade-off between mainlobe width (transition bandwidth $\Delta\omega$) and sidelobe attenuation (A_s) across window families.
5. **Evaluate** time-domain equations and spectral characteristics for Rectangular, Bartlett, Hann, Hamming, and Blackman windows.
6. **Synthesize** causal, linear-phase impulse responses $h_d[n]$ for Lowpass, Highpass, Bandpass, and Bandstop filters by incorporating delay $\tau = (N - 1)/2$.
7. **Execute** the end-to-end FIR filter design workflow to calculate numerical tap coefficients $h[n] = h_d[n] \cdot w[n]$.
8. **Verify** stopband attenuation and passband ripple tolerances against engineering specifications.

2. THEORETICAL FOUNDATIONS OF THE WINDOWING METHOD

2.1 The Inverse DTFT Dilemma and Windowing Concept

The design of an FIR filter begins with a desired continuous frequency response $H_d(e^{j\omega})$, which typically possesses an ideal “brickwall” magnitude characteristic:

$$H_{d,\text{LP}}(e^{j\omega}) = \begin{cases} 1 \cdot e^{-j\alpha\omega}, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases} \quad (1)$$

where α represents a desired constant linear-phase delay.

The corresponding ideal discrete-time impulse response $h_d[n]$ is obtained via the Inverse Discrete-Time Fourier Transform (IDTFT):

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\alpha\omega} e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} d\omega \quad (2)$$

Evaluating this elementary definite integral:

$$h_d[n] = \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)} \right]_{-\omega_c}^{\omega_c} = \frac{e^{j\omega_c(n-\alpha)} - e^{-j\omega_c(n-\alpha)}}{2\pi j(n-\alpha)} = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)} \quad (3)$$

At the exact center of symmetry $n = \alpha$, (3) yields an indeterminate 0/0 ratio. Applying L'Hôpital's rule with respect to n :

$$\lim_{n \rightarrow \alpha} h_d[n] = \lim_{n \rightarrow \alpha} \frac{\omega_c \cos(\omega_c(n-\alpha))}{\pi} = \frac{\omega_c}{\pi} \quad (4)$$

The Fundamental Implementation Hurdle: The ideal impulse response $h_d[n]$ in (3) is a **sinc sequence** that decays as $1/n$:

1. It extends infinitely in time from $n = -\infty$ to $n = +\infty$.

2. For any finite delay α , it contains non-zero samples for $n < 0$, making it **strictly non-causal**.

Physical realization requires an impulse response that is causal ($h[n] = 0$ for $n < 0$) and finite in length ($h[n] = 0$ for $n \geq N$). The simplest and most intuitive technique to achieve this is the **Windowing Method**: we truncate $h_d[n]$ by multiplying it with a finite-length window sequence $w[n]$ of length N :

$$\mathbf{h}[n] = \mathbf{h}_d[n] \cdot \mathbf{w}[n], \quad 0 \leq n \leq N-1 \quad (5)$$

where $w[n] = 0$ for $n < 0$ and $n \geq N$. To preserve linear phase, the window sequence $w[n]$ must be symmetric about the exact same delay center $\alpha = (N-1)/2$:

$$w[n] = w[N-1-n] \quad (6)$$

2.2 Frequency-Domain Mechanism: Periodic Convolution

By the modulation (multiplication) theorem of the DTFT, pointwise multiplication in the time domain corresponds to **periodic (circular) convolution** in the frequency domain:

$$\mathbf{H}(e^{j\omega}) = \frac{1}{2\pi} \mathbf{H}_d(e^{j\omega}) * \mathbf{W}(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathbf{H}_d(e^{j\theta}) \mathbf{W}(e^{j(\omega-\theta)}) d\theta \quad (7)$$

where $W(e^{j\omega}) = \sum_{n=0}^{N-1} w[n]e^{-j\omega n}$ is the Fourier transform of the window.

Equation (7) is the cornerstone of window design:

- If the window were infinitely long, $W(e^{j\omega})$ would be an impulse $\delta(\omega)$, and $H(e^{j\omega})$ would match $H_d(e^{j\omega})$ perfectly.
- Because $w[n]$ is finite, $W(e^{j\omega})$ possesses a central peak called the **mainlobe** and oscillatory ripples called **sidelobes**.
- When the ideal brickwall discontinuity is convolved with the mainlobe of $W(e^{j\omega})$, the vertical step is smoothed into a continuous transition band of finite slope ($\Delta\omega$).
- When the step discontinuity is convolved with the sidelobes of $W(e^{j\omega})$, ripple oscillations (ringing) are injected into both the passband and stopband.

The Windowing Method: Time-Domain Truncation and the Gibbs Phenomenon

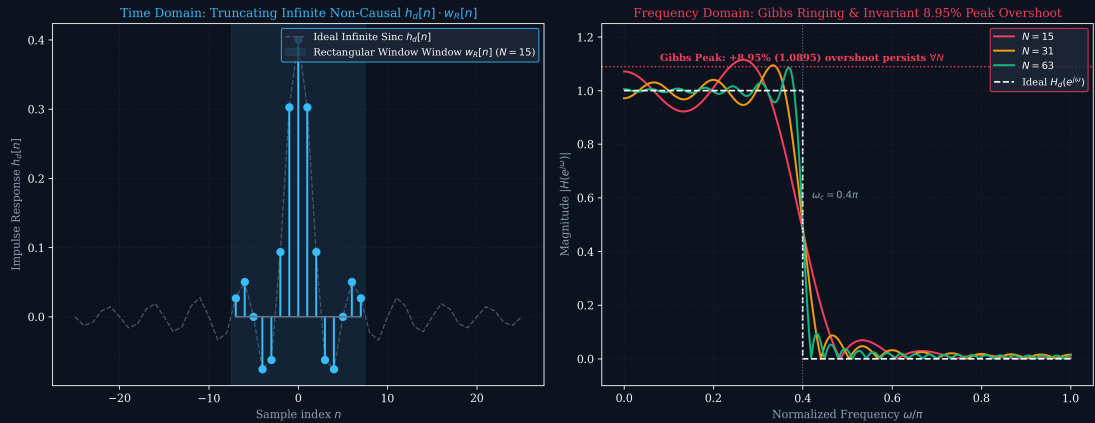


Figure 1: The Windowing Method and Gibbs Phenomenon: (Left) Time-domain truncation of the infinite sinc sequence $h_d[n]$ using a rectangular window ($N = 15$). (Right) Resulting frequency response $|H(e^{j\omega})|$ illustrating the invariant 8.95% peak overshoot (Gibbs ringing) at the band edge $\omega_c = 0.4\pi$ across various filter lengths $N = 15, 31, 63$.

2.3 The Gibbs Phenomenon

When a rectangular window $w_R[n] = 1$ ($0 \leq n \leq N - 1$) is applied, the time-domain impulse response is abruptly cut off. The frequency transform of the rectangular window centered at $n = 0$ is the Dirichlet kernel:

$$W_R(e^{j\omega}) = \sum_{n=-(N-1)/2}^{(N-1)/2} e^{-j\omega n} = \frac{\sin(\omega N/2)}{\sin(\omega/2)} \quad (8)$$

The first sidelobe of $W_R(e^{j\omega})$ occurs at approximately $\omega \approx 3\pi/N$, with a height of:

$$\frac{|W_R(3\pi/N)|}{|W_R(0)|} \approx \frac{2}{3\pi} \approx 0.212 \implies 20 \log_{10}(0.212) \approx -13.4 \text{ dB} \quad (9)$$

Convolving the ideal step discontinuity of $H_d(e^{j\omega})$ with this Dirichlet kernel produces the famous **Gibbs Phenomenon**:

1. An overshoot occurs directly adjacent to the discontinuity ω_c , reaching:

$$|H(e^{j\omega_{\text{peak}}})| = \frac{1}{2} + \frac{1}{\pi} \int_0^\pi \frac{\sin x}{x} dx = \frac{1}{2} + \frac{1}{\pi} \text{Si}(\pi) \approx 0.5 + 0.5895 = \mathbf{1.0895} \quad (10)$$

This corresponds to a **fixed 8.95% overshoot** (≈ 0.75 dB peak ripple) in the passband, and a minimum stopband attenuation of only $A_s = 21$ dB.

2. **The Invariance of the Peak Overshoot:** As filter length $N \rightarrow \infty$, the mainlobe narrows, compressing the ripple closer to the discontinuity. However, **the amplitude of the peak overshoot does NOT decrease!** It remains locked at 8.95%.
3. Therefore, a rectangular window can **never** be used for applications requiring stopband attenuation greater than 21 dB, regardless of how large N is chosen!

3. CLASSIC WINDOW FAMILIES AND PERFORMANCE TRADE-OFFS

To suppress Gibbs ringing and achieve high stopband attenuation, the window sequence must **taper smoothly to zero** at its boundaries $n = 0$ and $n = N - 1$. Eliminating the abrupt step discontinuity in time eliminates the high-frequency sidelobes in frequency.

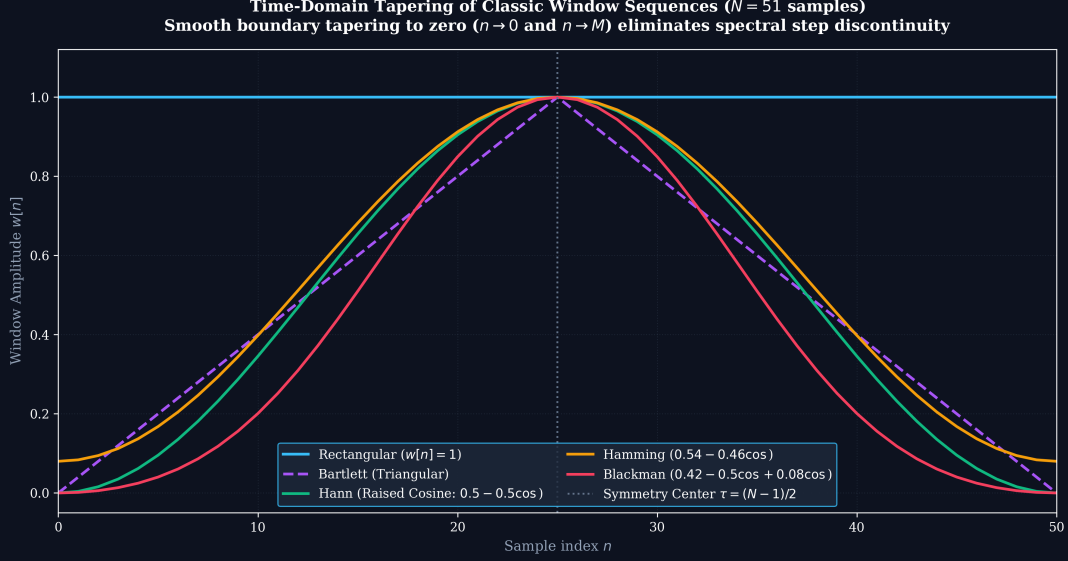


Figure 2: Time-Domain Profiles of Classic Window Sequences ($N = 51$): Comparing Rectangular, Bartlett, Hann, Hamming, and Blackman windows. Note the smooth tapering to near zero at the edges.

3.1 Bartlett (Triangular) Window

The Bartlett window linearly tapers from 0 at the boundaries to 1 at the center:

$$w_B[n] = 1 - \frac{2 \left| n - \frac{N-1}{2} \right|}{N-1}, \quad 0 \leq n \leq N-1 \quad (11)$$

In the frequency domain, the Bartlett window is the convolution of two rectangular windows of length $N/2$, meaning its spectrum is proportional to $[\sin(\omega N/4)/\sin(\omega/2)]^2$. Because the spectrum is squared, its sidelobes decay as $1/\omega^2$ rather than $1/\omega$:

- Mainlobe Width: $\Delta\omega = \frac{8\pi}{N}$ (doubled compared to rectangular).
- Peak Sidelobe Level: **-25 dB**.
- Minimum Stopband Attenuation: **$A_s = 25$ dB**.

3.2 Hann (Hanning) Window (Raised Cosine)

The Hann window applies a full period of a raised cosine function:

$$w_{Hn}[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) = \sin^2\left(\frac{\pi n}{N-1}\right), \quad 0 \leq n \leq N-1 \quad (12)$$

Notice that $w_{Hn}[0] = w_{Hn}[N-1] = 0$. Both the window and its first derivative reach zero smoothly at the endpoints:

- Mainlobe Width: $\Delta\omega = \frac{8\pi}{N}$.
- Peak Sidelobe Level: **-31 dB**.
- Sidelobe Roll-off Rate: **-18 dB/octave** (decays rapidly as $1/\omega^3$).
- Minimum Stopband Attenuation: **$A_s = 44$ dB**.

3.3 3.3 Hamming Window (Shifted Raised Cosine)

The Hamming window is an optimized modification of the raised cosine that adds a small pedestal to cancel the first and most destructive sidelobe:

$$w_{Hm}[n] = \alpha - (1 - \alpha) \cos\left(\frac{2\pi n}{N-1}\right) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \leq n \leq N-1 \quad (13)$$

By setting $\alpha = 0.54$, the positive first sidelobe of the rectangular component exactly cancels the negative sidelobe of the cosine component:

- Mainlobe Width: $\Delta\omega = \frac{8\pi}{N}$.
- Peak Sidelobe Level: -41 dB (a 10 dB improvement over Hann!).
- Minimum Stopband Attenuation: $\mathbf{A_s = 53}$ dB.
- **Trade-off vs. Hann:** Because $w_{Hm}[0] = 0.08 \neq 0$, its boundary is discontinuous, so far-out sidelobes roll off at only -6 dB/octave (slower than Hann). However, its peak near-in sidelobe is significantly lower.

3.4 3.4 Blackman Window (Three-Term Cosine Series)

The Blackman window introduces a second harmonic cosine term to achieve extreme sidelobe suppression:

$$w_{Bl}[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right), \quad 0 \leq n \leq N-1 \quad (14)$$

- Mainlobe Width: $\Delta\omega = \frac{12\pi}{N}$ (three times wider than rectangular).
- Peak Sidelobe Level: -57 dB.
- Minimum Stopband Attenuation: $\mathbf{A_s = 74}$ dB (suitable for high-fidelity audio and 12-bit ADC DSP front-ends).

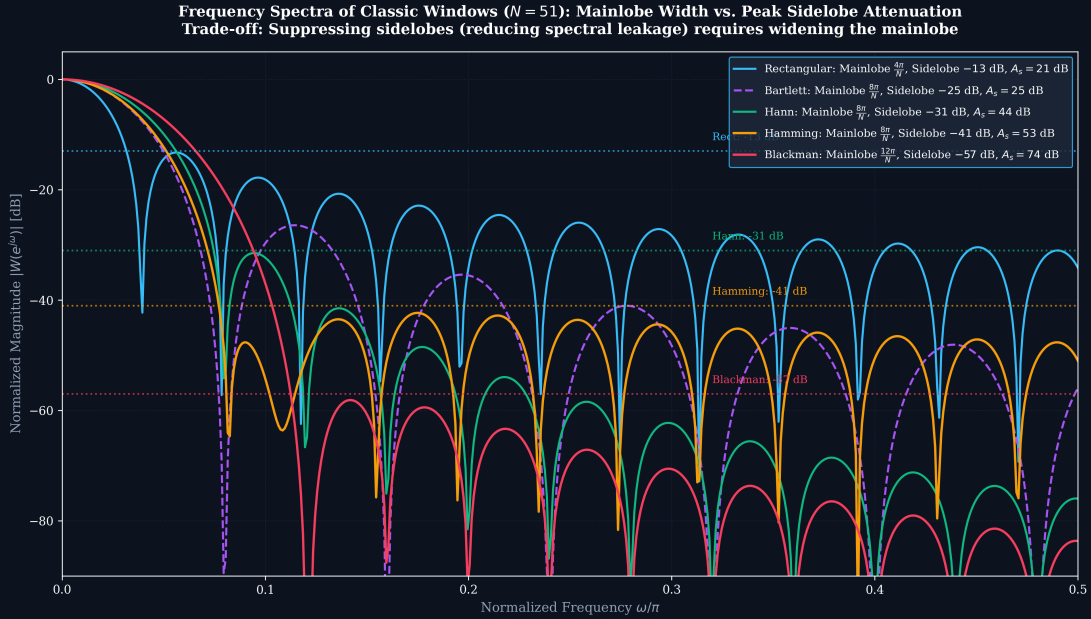


Figure 3: Frequency Magnitude Spectra of Classic Windows ($N = 51$): Comparing mainlobe widths and peak sidelobe rejection levels (-13 dB for Rectangular, -25 dB for Bartlett, -31 dB for Hann, -41 dB for Hamming, and -57 dB for Blackman).

3.5 3.5 Comprehensive Window Comparison Audit

Table 1 summarizes the quantitative performance metrics of the standard window families.

Table 1: Quantitative Performance Audit of Standard FIR Window Functions

Window Type	Time-Domain Definition $w[n]$ ($0 \leq n \leq N-1$)	Mainlobe Width	Peak Sidelobe	Min. Attenuation A_s	Transition Width $\Delta f = \frac{\Delta\omega}{2\pi}$
Rectangular	1	$4\pi/N$	-13.3 dB	21 dB	$\approx 0.9/N$
Bartlett	$1 - \frac{2 n-(N-1)/2 }{N-1}$	$8\pi/N$	-25.0 dB	25 dB	$\approx 1.6/N$
Hann	$0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right)$	$8\pi/N$	-31.5 dB	44 dB	$\approx 3.1/N$
Hamming	$0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$	$8\pi/N$	-41.0 dB	53 dB	$\approx 3.3/N$
Blackman	$0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$	$12\pi/N$	-57.0 dB	74 dB	$\approx 5.5/N$

4 4. CAUSAL IDEAL IMPULSE RESPONSES FOR STANDARD FILTER TYPES

To synthesize causal linear-phase filters, the desired frequency response must incorporate the delay factor $e^{-j\tau\omega}$, where $\tau = (N-1)/2$. The analytical closed-form expressions for the four standard filter archetypes are derived below:

4.1 4.1 Ideal Lowpass Filter (LPF)

$$H_{d,\text{LP}}(e^{j\omega}) = \begin{cases} e^{-j\tau\omega}, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases} \implies \mathbf{h}_{d,\text{LP}}[\mathbf{n}] = \frac{\sin(\omega_c(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} \quad (15)$$

At $n = \tau$: $h_{d,\text{LP}}[\tau] = \frac{\omega_c}{\pi}$.

4.2 4.2 Ideal Highpass Filter (HPF)

Using the complementary relationship $H_{d,\text{HP}}(e^{j\omega}) = e^{-j\tau\omega} - H_{d,\text{LP}}(e^{j\omega})$:

$$\mathbf{h}_{d,\text{HP}}[\mathbf{n}] = \frac{\sin(\pi(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} - \frac{\sin(\omega_c(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} \quad (16)$$

Since $\frac{\sin(\pi(n-\tau))}{\pi(n-\tau)} = \delta[n-\tau]$ when τ is an integer:

$$h_{d,\text{HP}}[n] = \begin{cases} 1 - \frac{\omega_c}{\pi}, & n = \tau \\ -\frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}, & n \neq \tau \end{cases} \quad (17)$$

4.3 4.3 Ideal Bandpass Filter (BPF)

With lower cutoff ω_{c1} and upper cutoff ω_{c2} ($\omega_{c1} < \omega_{c2}$):

$$\mathbf{h}_{d,\text{BP}}[\mathbf{n}] = \frac{\sin(\omega_{c2}(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} - \frac{\sin(\omega_{c1}(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} \quad (18)$$

At $n = \tau$: $h_{d,\text{BP}}[\tau] = \frac{\omega_{c2} - \omega_{c1}}{\pi}$.

4.4 Ideal Bandstop Filter (BSF)

Complementary to the bandpass filter:

$$\mathbf{h}_{d,BS}[\mathbf{n}] = \frac{\sin(\omega_{c1}(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} + \left[\frac{\sin(\pi(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} - \frac{\sin(\omega_{c2}(\mathbf{n} - \tau))}{\pi(\mathbf{n} - \tau)} \right] \quad (19)$$

At $n = \tau$: $h_{d,BS}[\tau] = 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi}$.

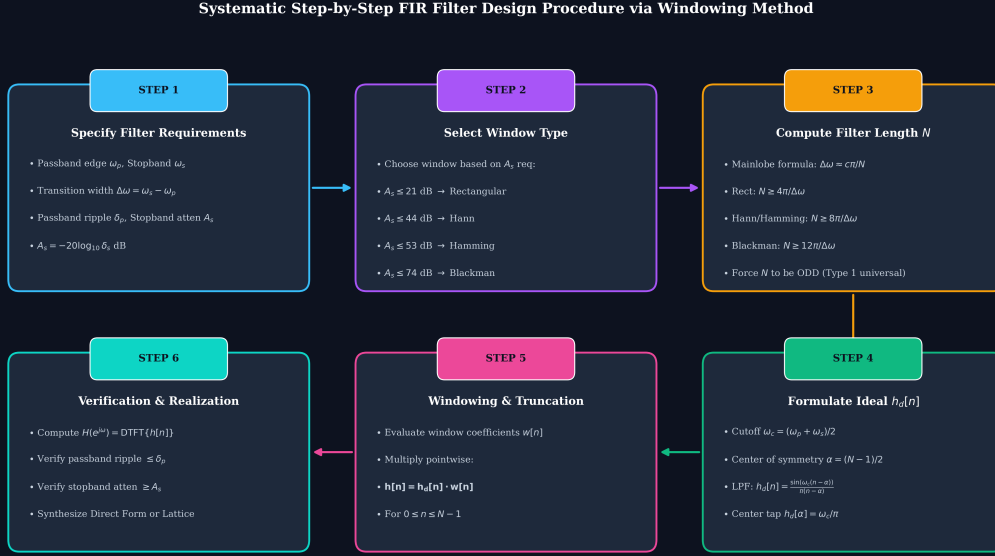


Figure 4: Systematic End-to-End FIR Filter Design Flowchart: Illustrating the 6 progressive design phases from frequency specifications to hardware-ready coefficients.

5. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

5.1 Example 22.1: Causal Lowpass Filter Design using Hann Window

Problem Statement: Design a 7-tap causal linear-phase FIR lowpass filter with a cutoff frequency $\omega_c = \pi/4$ radians/sample using a Hann window.

1. Determine the filter order M , length N , and delay center τ .
2. Calculate the exact analytical values of the ideal impulse response $h_d[n]$ for $n = 0, 1, \dots, 6$.
3. Compute the Hann window sequence $w[n]$ for $n = 0, 1, \dots, 6$.
4. Derive the final filter coefficients $h[n] = h_d[n] \cdot w[n]$.
5. Evaluate the DC gain $H(e^{j0})$ and Nyquist gain $H(e^{j\pi})$.

Solution:

Step 1: Filter Parameters

- Filter length: $N = 7$ (Odd $\implies$ Type 1 FIR Filter).
- Filter order: $M = N - 1 = 6$.

- Delay center: $\tau = \frac{N-1}{2} = \frac{7-1}{2} = \mathbf{3}$ samples (integer).

Step 2: Ideal Impulse Response Evaluation Using (15) with $\omega_c = \pi/4$ and $\tau = 3$:

$$h_d[n] = \frac{\sin\left(\frac{\pi}{4}(n-3)\right)}{\pi(n-3)} \quad (20)$$

Exploiting symmetry $h_d[3-k] = h_d[3+k]$:

- **Center tap ($n = 3$):** Using (4):

$$h_d[3] = \frac{\omega_c}{\pi} = \frac{\pi/4}{\pi} = \mathbf{0.2500} \quad (21)$$

- **Taps $n = 2$ and $n = 4$ ($k = 1$):**

$$h_d[2] = h_d[4] = \frac{\sin(-\pi/4)}{\pi(-1)} = \frac{-\sqrt{2}/2}{-\pi} = \frac{\sqrt{2}}{2\pi} \approx \frac{0.707107}{3.141593} = \mathbf{0.225079} \quad (22)$$

- **Taps $n = 1$ and $n = 5$ ($k = 2$):**

$$h_d[1] = h_d[5] = \frac{\sin(-2\pi/4)}{\pi(-2)} = \frac{\sin(-\pi/2)}{-2\pi} = \frac{-1}{-2\pi} = \frac{1}{2\pi} \approx \mathbf{0.159155} \quad (23)$$

- **Taps $n = 0$ and $n = 6$ ($k = 3$):**

$$h_d[0] = h_d[6] = \frac{\sin(-3\pi/4)}{\pi(-3)} = \frac{-\sqrt{2}/2}{-3\pi} = \frac{\sqrt{2}}{6\pi} \approx \frac{0.707107}{9.424778} = \mathbf{0.075026} \quad (24)$$

Step 3: Hann Window Sequence Evaluation Using (12) with $N = 7$:

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{6}\right) = 0.5 - 0.5 \cos\left(\frac{\pi n}{3}\right) \quad (25)$$

- $w[0] = w[6] = 0.5 - 0.5 \cos(0) = 0.5 - 0.5(1) = \mathbf{0.0000}$
- $w[1] = w[5] = 0.5 - 0.5 \cos(\pi/3) = 0.5 - 0.5(0.5) = 0.5 - 0.25 = \mathbf{0.2500}$
- $w[2] = w[4] = 0.5 - 0.5 \cos(2\pi/3) = 0.5 - 0.5(-0.5) = 0.5 + 0.25 = \mathbf{0.7500}$
- $w[3] = 0.5 - 0.5 \cos(\pi) = 0.5 - 0.5(-1) = 0.5 + 0.5 = \mathbf{1.0000}$

Step 4: Windowed FIR Tap Coefficients $h[n] = h_d[n] \cdot w[n]$

$$h[0] = h_d[0] \cdot w[0] = 0.075026 \times 0.0000 = \mathbf{0.0000} \quad (26)$$

$$h[1] = h_d[1] \cdot w[1] = 0.159155 \times 0.2500 = \mathbf{0.039789} \quad (27)$$

$$h[2] = h_d[2] \cdot w[2] = 0.225079 \times 0.7500 = \mathbf{0.168809} \quad (28)$$

$$h[3] = h_d[3] \cdot w[3] = 0.250000 \times 1.0000 = \mathbf{0.250000} \quad (29)$$

$$h[4] = h[2] = \mathbf{0.168809} \quad (30)$$

$$h[5] = h[1] = \mathbf{0.039789} \quad (31)$$

$$h[6] = h[0] = \mathbf{0.0000} \quad (32)$$

The synthesized causal impulse response is:

$$\mathbf{h[n] = \{0.0, 0.0398, 0.1688, 0.2500, 0.1688, 0.0398, 0.0\}} \quad (33)$$

Step 5: Gain Verification

- **DC Gain** ($\omega = 0$):

$$H(e^{j0}) = \sum_{n=0}^6 h[n] = 0.2500 + 2(0.168809) + 2(0.039789) = 0.2500 + 0.3376 + 0.0796 = \mathbf{0.6672} \quad (34)$$

(Normalized DC gain can be scaled to unity by dividing each coefficient by 0.6672).

- **Nyquist Gain** ($\omega = \pi$):

$$\begin{aligned} H(e^{j\pi}) &= \sum_{n=0}^6 h[n](-1)^n = h[0] - h[1] + h[2] - h[3] + h[4] - h[5] + h[6] \\ &= 0 - 0.0398 + 0.1688 - 0.2500 + 0.1688 - 0.0398 + 0 = \mathbf{0.0080} \end{aligned} \quad (35)$$

Stopband attenuation at Nyquist: $20 \log_{10}(0.0080/0.6672) = -\mathbf{38.4}$ dB! Even with only 7 taps, the Hann window achieves impressive stopband suppression.

5.2 Example 22.2: 9-Tap Highpass Filter Design (Rectangular vs. Hamming)

Problem Statement: Design a 9-tap linear-phase FIR Highpass filter with cutoff $\omega_c = \pi/2$ radians/sample using:

1. A Rectangular window.
2. A Hamming window.

Compare the resulting filter coefficients, boundary zeros, and discuss passband ripple behavior.

Solution:

Step 1: Filter Specifications

- Length $N = 9$ (Odd $\implies$ Type 1 FIR Filter).
- Delay center: $\tau = (9 - 1)/2 = 4$ samples.
- Cutoff: $\omega_c = \pi/2$.

Step 2: Ideal Highpass Impulse Response From (16):

$$h_{d,\text{HP}}[n] = \frac{\sin(\pi(n-4))}{\pi(n-4)} - \frac{\sin\left(\frac{\pi}{2}(n-4)\right)}{\pi(n-4)} \quad (36)$$

Evaluating for $n = 0, 1, 2, 3, 4$:

- Center tap ($n = 4$): $h_{d,\text{HP}}[4] = 1 - \frac{\pi/2}{\pi} = 1 - 0.5 = \mathbf{0.5000}$.
- Tap $n = 3$ ($n - 4 = -1$): $h_d[3] = 0 - \frac{\sin(-\pi/2)}{-\pi} = -\frac{-1}{-\pi} = -\frac{1}{\pi} \approx -\mathbf{0.318310}$.
- Tap $n = 2$ ($n - 4 = -2$): $h_d[2] = 0 - \frac{\sin(-\pi)}{-2\pi} = \mathbf{0.0000}$.
- Tap $n = 1$ ($n - 4 = -3$): $h_d[1] = 0 - \frac{\sin(-3\pi/2)}{-3\pi} = -\frac{1}{-3\pi} = \frac{1}{3\pi} \approx \mathbf{0.106103}$.
- Tap $n = 0$ ($n - 4 = -4$): $h_d[0] = 0 - \frac{\sin(-2\pi)}{-4\pi} = \mathbf{0.0000}$.

Step 3: Design with Rectangular Window ($w_R[n] = 1$) Since $w_R[n] = 1$, $h_R[n] = h_d[n]$:

$$\mathbf{h_R[n]} = \{0.0, 0.1061, 0.0, -0.3183, 0.5000, -0.3183, 0.0, 0.1061, 0.0\} \quad (37)$$

DC Gain: $H_R(e^{j0}) = 0.5 + 2(-0.3183) + 2(0.1061) = 0.5 - 0.6366 + 0.2122 = \mathbf{0.0756}$ (poor stopband rejection ≈ -16 dB due to Gibbs phenomenon).

Step 4: Design with Hamming Window Formula: $w_{Hm}[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{8}\right) = 0.54 - 0.46 \cos\left(\frac{\pi n}{4}\right)$:

$$w[4] = 0.54 - 0.46 \cos(\pi) = 0.54 + 0.46 = 1.0000 \quad (38)$$

$$w[3] = w[5] = 0.54 - 0.46 \cos(3\pi/4) = 0.54 - 0.46(-0.7071) = 0.54 + 0.3253 = 0.8653 \quad (39)$$

$$w[2] = w[6] = 0.54 - 0.46 \cos(\pi/2) = 0.54 - 0 = 0.5400 \quad (40)$$

$$w[1] = w[7] = 0.54 - 0.46 \cos(\pi/4) = 0.54 - 0.46(0.7071) = 0.54 - 0.3253 = 0.2147 \quad (41)$$

$$w[0] = w[8] = 0.54 - 0.46(1) = 0.0800 \quad (42)$$

Pointwise multiplication $h_{Hm}[n] = h_d[n] \cdot w[n]$:

$$h_{Hm}[4] = 0.5000 \times 1.0000 = \mathbf{0.5000} \quad (43)$$

$$h_{Hm}[3] = h_{Hm}[5] = -0.318310 \times 0.8653 = \mathbf{-0.2754} \quad (44)$$

$$h_{Hm}[2] = h_{Hm}[6] = 0.0000 \times 0.5400 = \mathbf{0.0000} \quad (45)$$

$$h_{Hm}[1] = h_{Hm}[7] = 0.106103 \times 0.2147 = \mathbf{0.0228} \quad (46)$$

$$h_{Hm}[0] = h_{Hm}[8] = 0.0000 \times 0.0800 = \mathbf{0.0000} \quad (47)$$

DC Gain: $H_{Hm}(e^{j0}) = 0.5 + 2(-0.2754) + 2(0.0228) = 0.5 - 0.5508 + 0.0456 = \mathbf{-0.0052!}$
 Stopband attenuation at DC: $20 \log_{10}(|-0.0052|) = \mathbf{-45.7}$ dB. The Hamming window provides almost 30 dB greater rejection than the rectangular window!

5.3 Example 22.3: 11-Tap Bandpass Filter Design using Bartlett Window

Problem Statement: Design an 11-tap causal FIR Bandpass filter with lower cutoff $\omega_{c1} = 0.3\pi$ and upper cutoff $\omega_{c2} = 0.6\pi$ using a Bartlett (triangular) window. Find all non-zero coefficients.

Solution:

Step 1: Parameters

- Length $N = 11$, Order $M = 10$, Center $\tau = (11 - 1)/2 = \mathbf{5}$.
- Passband bandwidth: $\Delta\omega_c = \omega_{c2} - \omega_{c1} = 0.6\pi - 0.3\pi = 0.3\pi$.

Step 2: Ideal Bandpass Impulse Response From (18):

$$h_d[n] = \frac{\sin(0.6\pi(n-5))}{\pi(n-5)} - \frac{\sin(0.3\pi(n-5))}{\pi(n-5)} \quad (48)$$

- $n = 5$: $h_d[5] = \frac{0.6\pi - 0.3\pi}{\pi} = \mathbf{0.3000}$.
- $n = 4$ ($k = -1$): $h_d[4] = \frac{\sin(-0.6\pi)}{-\pi} - \frac{\sin(-0.3\pi)}{-\pi} = \frac{-0.951057}{-\pi} - \frac{-0.809017}{-\pi} = 0.30273 - 0.25752 = \mathbf{0.04521}$.
- $n = 3$ ($k = -2$): $h_d[3] = \frac{\sin(-1.2\pi)}{-2\pi} - \frac{\sin(-0.6\pi)}{-2\pi} = \frac{0.587785}{-2\pi} - \frac{-0.951057}{-2\pi} = -0.09355 - 0.15137 = \mathbf{-0.24492}$.
- $n = 2$ ($k = -3$): $h_d[2] = \frac{\sin(-1.8\pi)}{-3\pi} - \frac{\sin(-0.9\pi)}{-3\pi} = \frac{0.587785}{-3\pi} - \frac{-0.309017}{-3\pi} = -0.06237 - 0.03279 = \mathbf{-0.09516}$.

- $n = 1$ ($k = -4$): $h_d[1] = \frac{\sin(-2.4\pi)}{-4\pi} - \frac{\sin(-1.2\pi)}{-4\pi} = \frac{-0.951057}{-4\pi} - \frac{0.587785}{-4\pi} = 0.07568 + 0.04677 = \mathbf{0.12245}$.
- $n = 0$ ($k = -5$): $h_d[0] = \frac{\sin(-3\pi)}{-5\pi} - \frac{\sin(-1.5\pi)}{-5\pi} = 0 - \frac{1}{-5\pi} = \frac{1}{5\pi} = \mathbf{0.06366}$.

Step 3: Bartlett Window Evaluation Using (11) with $M = 10$: $w_B[n] = 1 - \frac{|n-5|}{5}$:

$$w_B[5] = 1.0, \quad w_B[4] = w_B[6] = 0.8, \quad w_B[3] = w_B[7] = 0.6 \quad (49)$$

$$w_B[2] = w_B[8] = 0.4, \quad w_B[1] = w_B[9] = 0.2, \quad w_B[0] = w_B[10] = 0.0 \quad (50)$$

Step 4: Windowed Filter Coefficients $h[n] = h_d[n] \cdot w_B[n]$

$$h[5] = 0.3000 \times 1.0 = \mathbf{0.3000} \quad (51)$$

$$h[4] = h[6] = 0.04521 \times 0.8 = \mathbf{0.03617} \quad (52)$$

$$h[3] = h[7] = -0.24492 \times 0.6 = \mathbf{-0.14695} \quad (53)$$

$$h[2] = h[8] = -0.09516 \times 0.4 = \mathbf{-0.03806} \quad (54)$$

$$h[1] = h[9] = 0.12245 \times 0.2 = \mathbf{0.02449} \quad (55)$$

$$h[0] = h[10] = 0.06366 \times 0.0 = \mathbf{0.0000} \quad (56)$$

The synthesized 11-tap Bandpass filter is:

$$\mathbf{h[n] = \{0.0, 0.0245, -0.0381, -0.1470, 0.0362, 0.3000, 0.0362, -0.1470, -0.0381, 0.0245, 0.0\}} \quad (57)$$

6. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Attempting to Reduce Gibbs Overshoot by Increasing N with a Rectangular Window:** This is the classic student blunder. Increasing N makes the transition slope steeper, but **the peak ripple remains fixed at 8.95%**. To reduce ripple amplitude, one **MUST** change the window function (e.g. to Hann, Hamming, or Blackman).
- **Pitfall 2: Forgetting L'Hôpital's Rule at the Center Tap $n = \tau$:** When calculating $h_d[\tau]$, evaluating $\sin(0)/0$ numerically causes a divide-by-zero crash in code or undefined value on paper. Always write:

$$h_d[\tau] = \frac{\omega_c}{\pi} \quad (\text{for LPF}), \quad 1 - \frac{\omega_c}{\pi} \quad (\text{for HPF}), \quad \frac{\omega_{c2} - \omega_{c1}}{\pi} \quad (\text{for BPF}) \quad (58)$$

- **Pitfall 3: Transition Bandwidth Formula Mismatch:** The transition bandwidth $\Delta\omega$ in rad/sample must not be confused with normalized frequency $\Delta f = \Delta\omega/(2\pi)$. For Hamming, $\Delta\omega \approx 6.6\pi/N$ (often approximated as $8\pi/N$), which gives $\Delta f \approx 3.3/N$. Using $\Delta\omega$ instead of Δf in length formulas produces an order-of-magnitude error.
- **Pitfall 4: Window Endpoints Being Zero:** For the Hann window of length N , $w[0] = w[N-1] = 0$. This means the first and last taps of the filter will be exactly zero: $h[0] = h[N-1] = 0$. Students often think they made a math error. In reality, an N -tap Hann filter has only $N-2$ active non-zero taps!
- **Pitfall 5: Window Symmetry Parity:** When designing a Highpass filter via windowing, N **must be odd** (Type 1). If an even length N is chosen, the resulting Type 2 filter has an inescapable zero at $\omega = \pi$, completely canceling the highpass passband!

7 7. QUICK REVISION & SUMMARY CHEAT-SHEET

Table 2: FIR Filter Design via Windowing Method Quick Reference

Concept / Formula	Mathematical Definition / Operational Rule
Windowing Equation	$h[n] = h_d[n] \cdot w[n], \quad 0 \leq n \leq N-1$
Frequency Domain Dual	$H(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) * W(e^{j\omega})$ (Periodic Convolution)
Ideal LPF Impulse Response	$h_d[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}, \quad h_d[\tau] = \frac{\omega_c}{\pi}, \quad \tau = (N-1)/2$
Gibbs Phenomenon	Rectangular window produces fixed 8.95% overshoot (~ 0.75 dB ripple), $A_s = 21$ dB
Rectangular Window	$w[n] = 1 \implies \Delta\omega = 4\pi/N, \quad \text{Peak Sidelobe} = -13 \text{ dB}, \quad A_s = 21 \text{ dB}$
Bartlett Window	$w[n] = 1 - \frac{2 n-\tau }{N-1} \implies \Delta\omega = 8\pi/N, \quad \text{Peak Sidelobe} = -25 \text{ dB}, \quad A_s = 25 \text{ dB}$
Hann Window	$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) \implies \Delta\omega = 8\pi/N, \quad \text{Peak Sidelobe} = -31 \text{ dB}, \quad A_s = 44 \text{ dB}$
Hamming Window	$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \implies \Delta\omega = 8\pi/N, \quad \text{Peak Sidelobe} = -41 \text{ dB}, \quad A_s = 53 \text{ dB}$
Blackman Window	$w[n] = 0.42 - 0.5 \cos + 0.08 \cos \implies \Delta\omega = 12\pi/N, \quad \text{Peak Sidelobe} = -57 \text{ dB}, \quad A_s = 74 \text{ dB}$
Length Calculation	$N \geq \frac{C}{\Delta f} = \frac{2\pi C}{\Delta\omega}$ (where $C = 0.9$ for Rect, 3.3 for Hamming, 5.5 for Blackman)
Universal Parity Rule	Always select N ODD (Type 1) to support all filter types without boundary zero restrictions